

Recap on Newton's 3 Laws of Motion

SS ILP Physics (Intermediate), Week 3 Class 2



Outline

- Recap on Newton's three Laws of Motion
- Practice questions

Newton's 1st Law of Motion

- An object remains its state of rest or uniform motion at a constant velocity unless it is acted upon a non-zero resultant force.

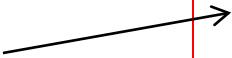
$$\sum \vec{F}_i = \vec{0} \rightarrow \vec{v} = \text{constant}$$

<https://www.physicsclassroom.com/class/newtlaws/Lesson-1/Newton-s-First-Law>

Newton's Second Law

- The acceleration is proportional to the resultant force, in the same direction and inversely proportional to the mass,

*Sum of all the
forces acting on that
object


$$\sum \vec{F}_i = \frac{d(m\vec{v})}{dt} = m\vec{a}$$

<https://www.physicsclassroom.com/class/newtlaws/Lesson-3/Newton-s-Second-Law>

Newton's Third Law

- To every action, there is always an opposite reaction of the same magnitude,

$$\vec{F}_{2,1} = -\vec{F}_{1,2}$$

force on object 2 due to object 1

force on object 1 due to object 2

<https://www.physicsclassroom.com/class/newtlaws/Lesson-4/Newton-s-Third-Law>

Solving dynamics problems with Newton's three laws

- I. Identify the object(s) of interest.
- II. Draw free body diagram(s) of the object(s).
- III. Choose a proper coordinate system and determine the components of the forces and acceleration along coordinate axes.
- IV. Apply Newton's 2nd law of motion to write equations with unknowns,

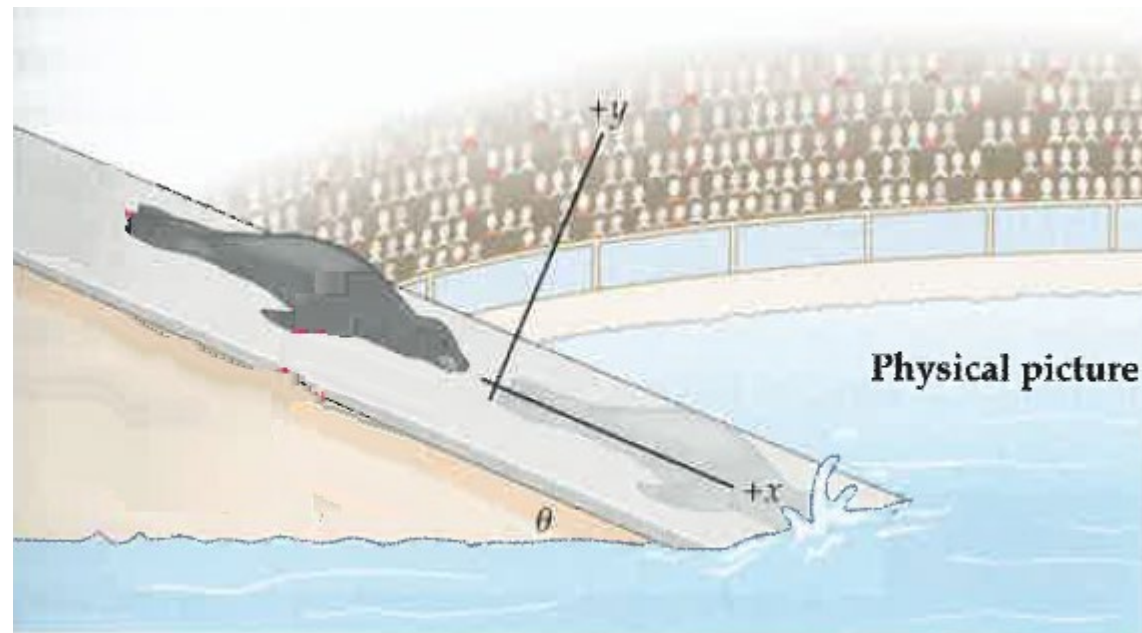
$$\sum_i^N \vec{F}_i = m\vec{a}$$

- V. In addition to Newton's 2nd law, identify any other equations you might need for solving unknowns.

H. D. Young and R. A. Freedman, University Physics (14th edition), p160.

Case problem 1

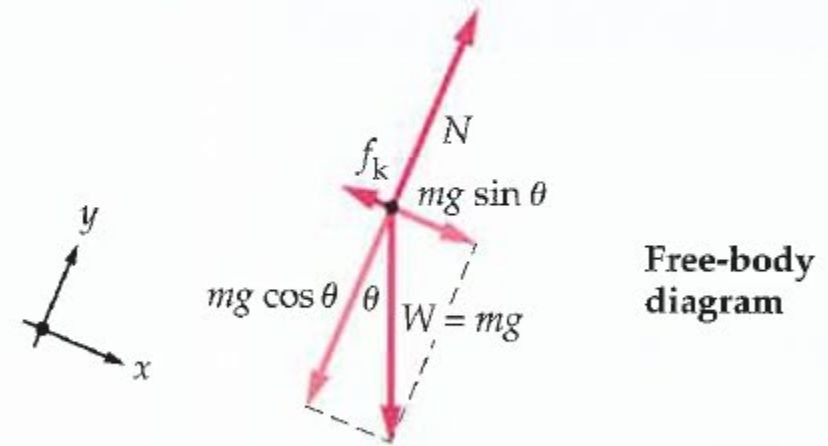
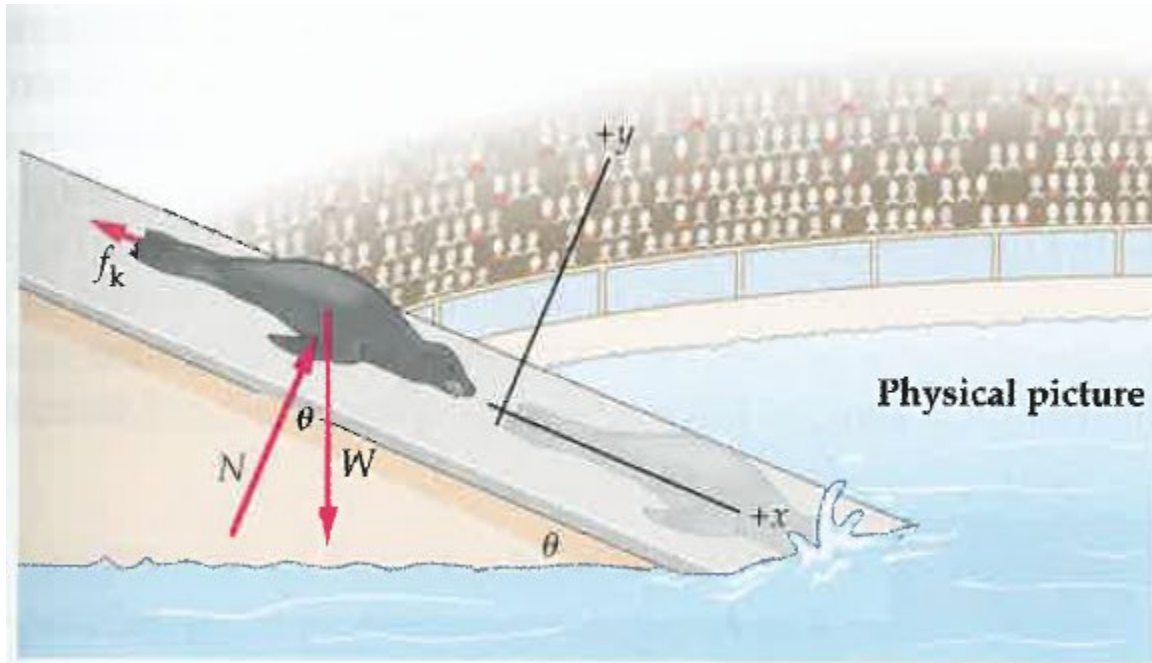
A trained sea lion slides from rest with constant acceleration down a 3.0m-long ramp into a pool of water. If the ramp is inclined at an angle of 23° above the horizontal and the coefficient of kinetic friction between the sea lion and the ramp is 0.26, how long does it take for the sea lion to make a splash in the pool? Take $g = 9.81 \text{ m/s}^2$.



J. S. Walker, Physics, Example 6-2, p145

Case problem 1: Solution

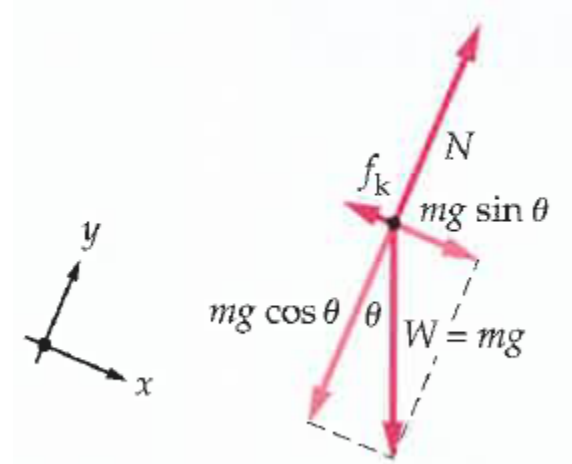
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$$\sum F_x = ma_x \rightarrow mg \sin \theta - \mu_k N = ma_x \rightarrow a_x = g(\sin \theta - \mu_k \cos \theta) = 1.5 \text{ m/s}^2$$

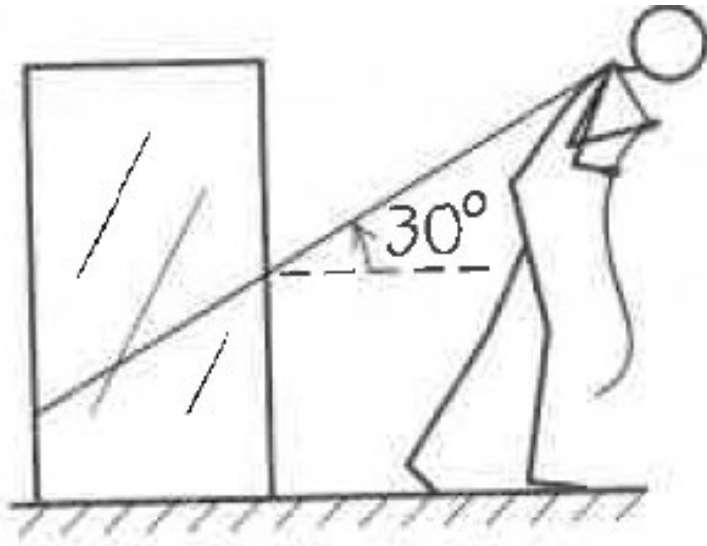
$$\sum F_y = ma_y \rightarrow N - mg \cos \theta = 0 \rightarrow N = mg \cos \theta$$

$$x = x_0 + v_{x,0}t + \frac{1}{2}a_x t^2 = \frac{1}{2}a_x t^2 \rightarrow t = \sqrt{\frac{2x}{a_x}} = 2.0 \text{ s}$$

J. S. Walker, Physics, Example 6-2, p145

Case problem 2

As shown in below figure, suppose you move the crate by pulling upward on the rope at an angle of 30° above the horizontal. How hard must you pull to keep it moving with constant velocity? Assume that $mg = 500\text{N}$ and $\mu_k = 0.4$.



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 5.15, p170

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$$\sum F_x = ma_x \rightarrow T \cos 30^\circ - f_k = 0 \rightarrow T \cos 30^\circ = f_k = \mu_k N$$

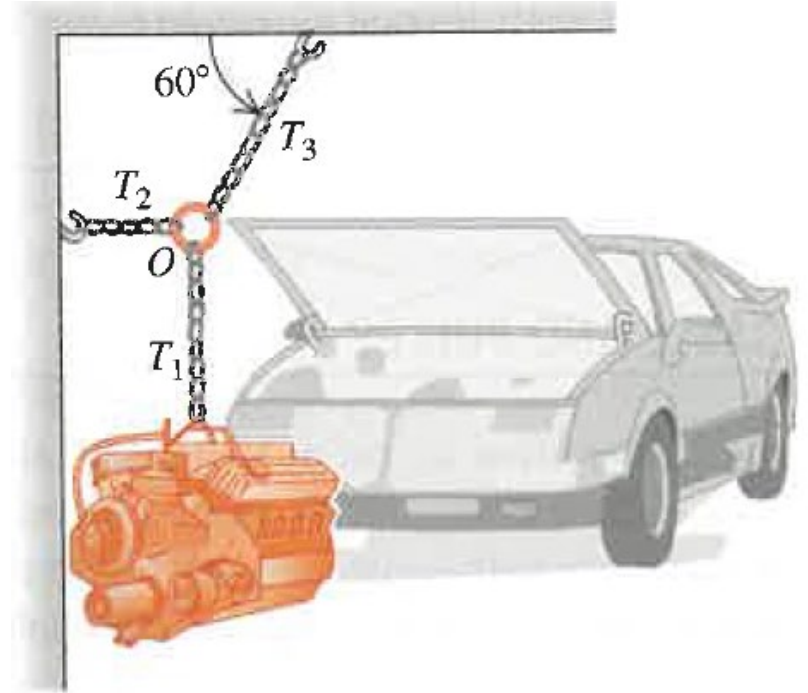
$$\sum F_y = ma_y \rightarrow T \sin 30^\circ + N - mg = 0 \rightarrow N = mg - T \sin 30^\circ$$

$$T \cos 30^\circ = \mu_k N = \mu_k (mg - T \sin 30^\circ) \rightarrow T = \frac{\mu_k mg}{\cos 30^\circ + \mu_k \sin 30^\circ} = 188 \text{ N}$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 5.15, p170

Case problem 3

A car engine with mass m hangs from a chain that is linked at ring O to two other chains, one fastened to the ceiling and the other to the wall. Find expression for the tension in each of the three chains in terms of m . The masses of the ring and chains are negligible compared with the mass of the engine and g .



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 5.3, p157

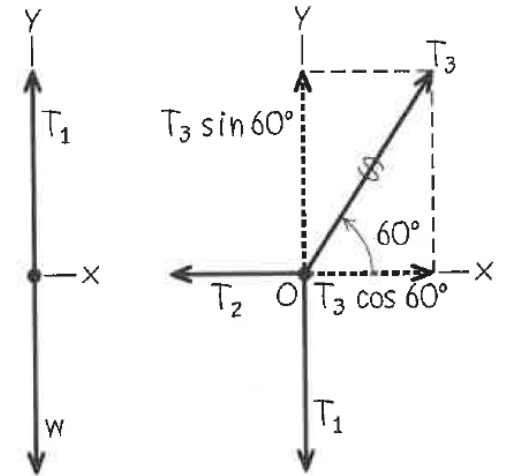
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$$\sum F_x = ma_x \rightarrow T_3 \cos 60^\circ - T_2 = 0 \rightarrow T_2 = T_3 \cos 60^\circ = \frac{mg \cos 60^\circ}{\sin 60^\circ}$$

$$\sum F_y = ma_y \rightarrow T_3 \sin 60^\circ - T_1 = 0 \rightarrow T_3 = \frac{T_1}{\sin 60^\circ} = \frac{mg}{\sin 60^\circ}$$

$$\sum F_y = ma_y \rightarrow T_1 = mg$$



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 5.3, p157

Case problem 4

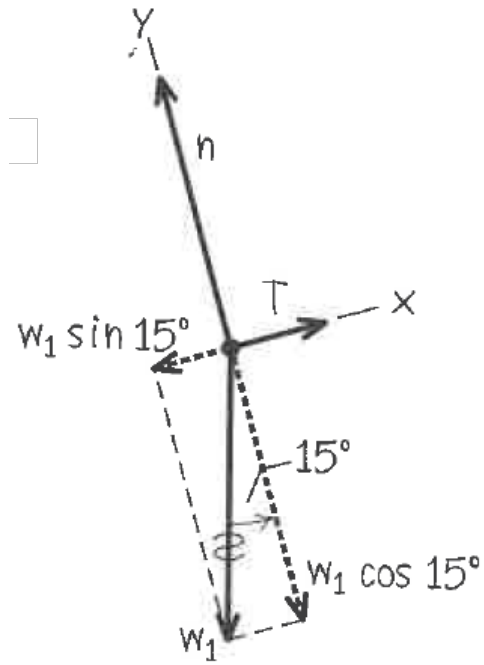
Your firm needs to haul granite blocks up to 15° slope out of a quarry and to lower dirt into the quarry to fill the holes. You design a system in which a granite block on a cart with steel wheels (mass m_1 , including both block and cart) is pulled uphill on steel rails by a dirt-filled bucket (mass m_2 , including both dirt and bucket) that descends vertically into the quarry. How must the masses m_1 and m_2 be related in order for the system to move with constant speed? Ignore friction in the pulley and wheels, and ignore the mass of the cable.



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 5.5, p158

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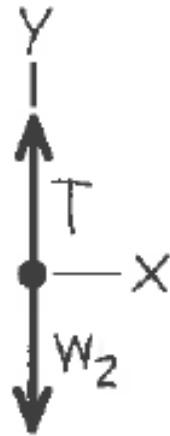


$$\begin{aligned} \sum F_x &= ma_x & \sum F_y &= ma_y \\ T - m_1 g \sin 15^\circ &= 0 & N - m_1 g \cos 15^\circ &= 0 \end{aligned}$$

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$$\sum F_x = ma_x$$

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$$T - m_2g = 0$$

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$$T - m_1 g \sin 15^\circ = 0 \quad N - m_1 g \cos 15^\circ = 0$$

$$\sum F_x = ma_x$$

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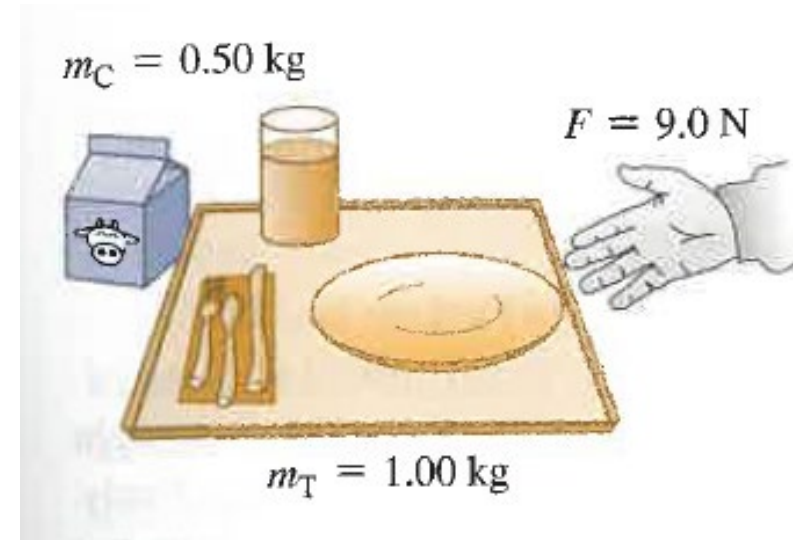
$$T - m_2 g = 0$$

$$m_1 g \sin 15^\circ = m_2 g \quad \rightarrow \quad m_1 \sin 15^\circ = m_2$$

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Case problem 5

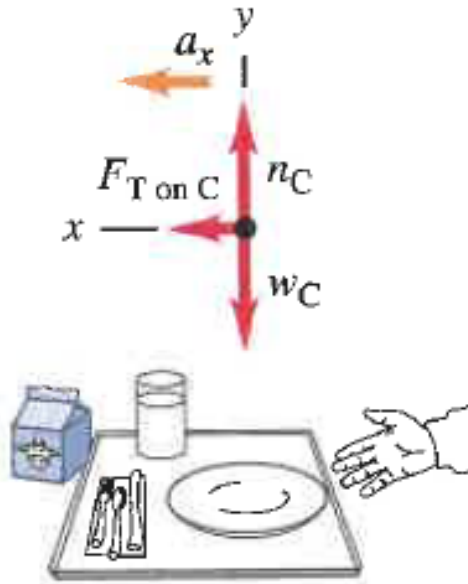
You push a 1.00-kg food tray through the cafeteria line with a constant 9.0-N force. The tray pushes a 0.50-kg milk carton. The tray and carton slide on a horizontal surface so greasy that friction can be ignored. Find the acceleration of the tray and carton and the horizontal force that the tray exerts on the carton.



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 5.11, p164

Case problem 5: Solution

You push a 1.00-kg food tray through the cafeteria line with a constant 9.0-N force. The tray pushes a 0.50-kg milk carton. The tray and carton slide on a horizontal surface so greasy that friction can be ignored. Find the acceleration of the tray and carton and the horizontal force that the tray exerts on the carton.



$$\sum F_x = ma_x$$

$$\sum F_y = ma_y$$

$$F_{T \text{ on } C} = m_C a_{Cx} \quad N_C - m_C g = m_C a_{Cy} = 0$$

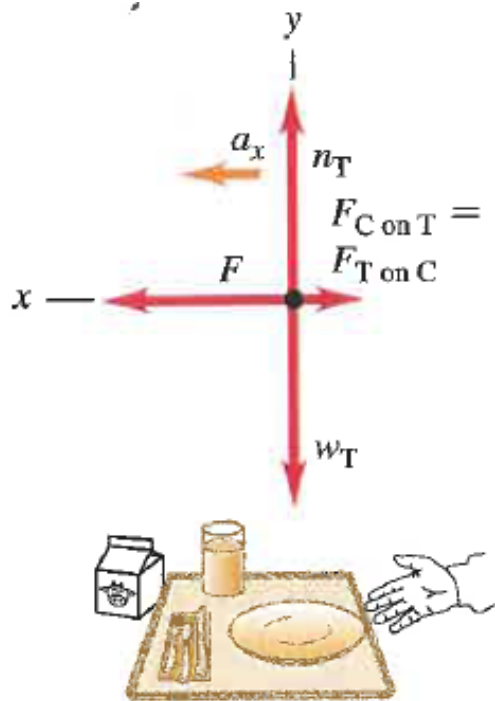
$$F_{T \text{ on } C} = m_C a$$

$$N_C = m_C g$$

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$$\sum F_x = ma_x$$

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$$F - F_{C \text{ on } T} = m_T a_{Tx} \quad N_T - m_T g = m_T a_{Ty} = 0$$

$$F - F_{C \text{ on } T} = m_T a$$

$$N_T = m_T g$$

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$$\sum F_x = ma_x$$

$$F_{T \text{ on } C} = m_C a_{Cx}$$

$$F_{T \text{ on } C} = m_C a$$

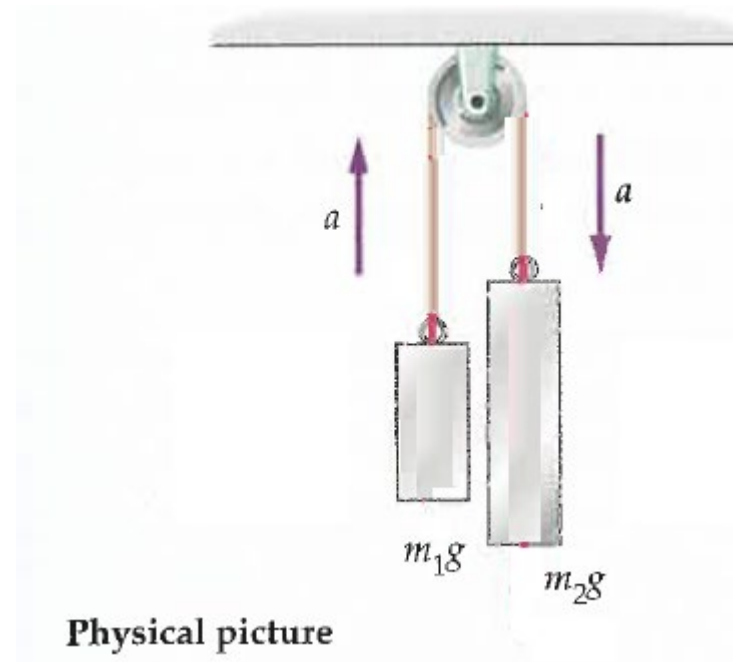
$$F - F_{C \text{ on } T} + F_{T \text{ on } C} = (m_C + m_T)a$$

$$\rightarrow a = \frac{F}{m_C + m_T} = 6 \text{ m/s}^2$$

$$F_{T \text{ on } C} = m_C a = \frac{m_C F}{m_C + m_T} = 3 \text{ N}$$

Case problem 6

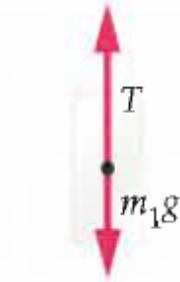
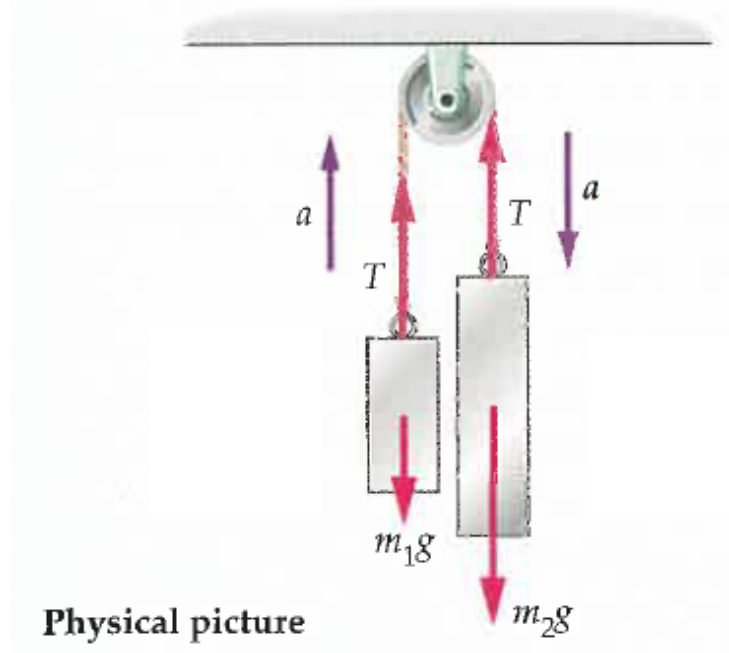
Atwood's machine consists of two masses connected by a string that passes over a pulley. Find the acceleration of the masses for general m_1 and m_2 , and evaluate for the specific case $m_1 = 3.1 \text{ kg}$ and $m_2 = 4.4 \text{ kg}$. Take $g = 9.81 \text{ m/s}^2$.



J. S. Walker, Physics, Example 6-7, p161

Case problem 6: Solution

Atwood's machine consists of two masses connected by a string that passes over a pulley. Find the acceleration of the masses for general m_1 and m_2 , and evaluate for the specific case $m_1 = 3.1 \text{ kg}$ and $m_2 = 4.4 \text{ kg}$. Take $g = 9.81 \text{ m/s}^2$.



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Atwood's machine consists of two masses connected by a string that passes over a pulley. Find the acceleration of the masses for general m_1 and m_2 , and evaluate for the specific case $m_1 = 3.1 \text{ kg}$ and $m_2 = 4.4 \text{ kg}$. Take $g = 9.81 \text{ m/s}^2$.

$$\begin{aligned}\sum F_y = ma_y &\rightarrow T - m_1g = m_1a_1 \\ &\rightarrow T - m_2g = -m_2a_2 \\ a_1 &= a_2 = a\end{aligned}$$

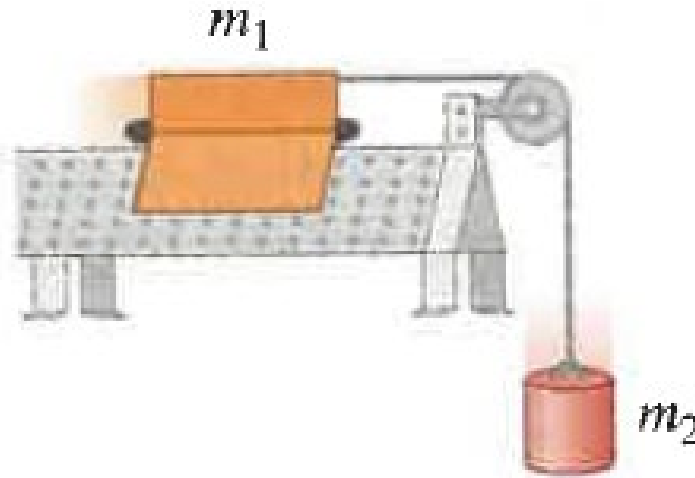
$$a = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g$$

$$a = \left(\frac{4.4 - 3.1}{3.1 + 4.4} \right) g = 1.7 \text{ m/s}^2$$

Off-class practice question(s)

Case problem

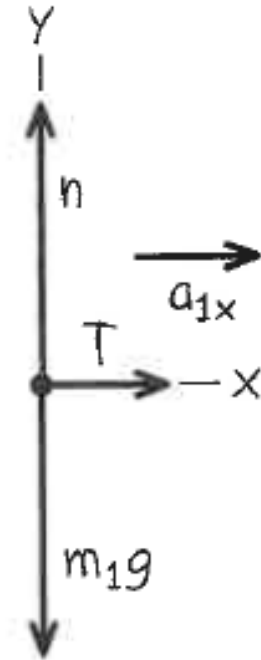
Below figure shows an air-track glider with mass m_1 moving on a level, frictionless air track in the physics lab. The glider is connected to a lab weight with mass m_2 by a light, flexible, non-stretching string that passes over a stationary, frictionless pulley. Find the acceleration of each body and the tension in the string.



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 5.12, p165

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$$T = m_1 a_{1x}$$

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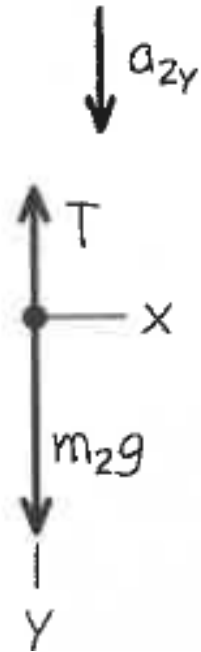
$$T = m_1 a$$

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$$m_2g - T = m_2a_{2y}$$

$$m_2g - T = m_2a$$

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$$T = m_1 a$$

$$N = m_1 g$$

$$\sum F_x = ma_x$$

$$\sum F_y = ma_y$$

$$m_2 g - T = m_2 a_{2y}$$

$$m_2 g - T = m_2 a$$

$$m_2 g = (m_1 + m_2) a \quad \rightarrow \quad a = \frac{m_2 g}{m_1 + m_2}$$

$$T = m_1 a = \frac{m_1 m_2 g}{m_1 + m_2}$$

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